

ELEC4631 Lecture 14: State-Feedback Control Using Lyapunov Functions

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Lecture outline

- Controlled ODEs.
- Lyapunov functions revisited.
- State-feedback design using Lyapunov functions.
- Controller design examples in motion control and mechanical systems.

Nonlinear controlled dynamical systems

- So far the course has focused on LTI systems with linear control schemes
- In this lecture we now consider nonlinear controlled dynamical systems represented by controlled nonlinear ODEs of the form:

$$\dot{x}(t) = f(x(t), u(t)); x(0) = x_0.$$

Here, $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, n is the dimension of the state-space ($S = \mathbb{R}^n$), and m is the dimension of the control signal $u(t)$, where $f(x, u)$ is no longer linear of the form $f(x, u) = Ax + Bu$.

- To control such systems in this lecture we will look at nonlinear control methods based on Lyapunov functions

State-feedback

- If we have direct access to the state $\boldsymbol{x}(t)$ at each time t (e.g., the state can be measurement directly) then a candidate control signal is one that is a function of the state:

$$u(t) = \eta(\boldsymbol{x}(t)).$$

Here, η is a function $\eta : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

- In the nonlinear context, the function η need not be a linear function of $\boldsymbol{x}$. Control using a signal of this form is a general form of state-feedback control.

State-feedback

- With state-feedback we have that:

$$\begin{aligned}\dot{x}(t) &= f(x(t), \eta(x(t))) \\ &= g(x(t)),\end{aligned}$$

where $g(x) = f(x, \eta(x))$.

- Thus, state-feedback reduces the controlled ODE to an ordinary ODE.
- With feedback control the loop is closed so we have a closed-loop system. Here g is the vector field of the closed-loop system.

Design using Lyapunov functions

- The problem, or control objective, to be considered is to drive the state $x(t)$ to a desired target set C in the state-space as $t \rightarrow \infty$. That is, $\lim_{t \rightarrow \infty} x(t) \in C$.
- Main idea in Lyapunov based control:

Simultaneously design a Lyapunov function $V(x)$ and the control function $\eta(x)$ to ensure that the state gets driven to the target set C .

Recalling Lyapunov functions

- Recall that V is a Lyapunov function around an equilibrium point x^* if it has the properties:
 1. V is lpd around x^* .
 2. $\dot{V}$ is lnsd or lnd around x^* .
- Based on properties of the Lyapunov function constructed we can infer the type of stability of the equilibrium point.

Extending the Lyapunov function

- As stated earlier, now we also want to be able to drive the state to a set C that may contain many points, rather than just to a single point like an equilibrium point x^* .
- For instance, in formation control of multiple vehicles (e.g., a platoon of unmanned aircrafts) we may want them, for instance, to maintain a formation shape (e.g., a circle).
- To accomodate for these more general control tasks, we will extend the properties required of the Lyapunov functions.
- Two types of target sets: A target set C is bounded if there is an $R > 0$ such that $\|x\| \leq R$ for all elements $x \in C$. Otherwise, C is unbounded.

Extending the Lyapunov function

- Take the state-space to be $S = \mathbb{R}^n$. For the purpose of control, we modify the requirements for a Lyapunov function as follows.
 1. $V(x) = 0$ for all x in a target set C in S and $V(x) > 0$ for all $x \notin C$.
 2. $\dot{V}(x) = 0$ for all $x \in C$.
 3. $\dot{V}(x) \leq 0$ for all $x \notin C$.

Extending the Lyapunov function

- Example: $V(x) = (x - v)^T P_1 (x - v)$ for a real symmetric positive definite matrix P_1 . Then $V(x) = 0$ if and only for $x = v$. Thus, here C only contains a single point: $C = \{v\}$. C is bounded.
- Example: Let $x = (x_1, x_2)$ and $V(x) = (x_1^2 + x_2^2 - r^2)^2$. Then $V(x) = 0$ if and only if x lies on the circle $\{(x_1, x_2) \mid x_1^2 + x_2^2 = r^2\}$. The target set C is this circle. C is bounded.
- Example: $V(x) = (x - v)^T P_2 (x - v)$ for a real symmetric positive semidefinite matrix P_2 . Then $V(x) = 0$ only for $x - v \in \text{null}(P)$. Thus, here C is a hyperplane passing through v : $C = \{v + z \mid z \in \text{null}(P)\}$. C is unbounded.

The main idea

- Find an (ideal) extended Lyapunov function V and a state-feedback $u(t) = \eta(x(t))$ such that
 1. $V(x) = 0$ for all $x \in C$, and $V(x) > 0$ for all $x \notin C$.
 2. $\dot{V}(x) = 0$ for all $x \in C$.
 3. $\dot{V}(x) = \nabla V(x)^T f(x, \eta(x)) < 0$ for all $x \notin C$.
 4. $\lim_{\|x\| \rightarrow \infty} V(x) = \infty$ if C is bounded.

- By requirement 3, $u(t) = \eta(x(t))$ will be such that $V(x(t)) \downarrow 0$ as $t \rightarrow \infty$.
- By requirements 1 and 2, when $x(t)$ is in C it stays in that set.
- Requirement 4 is the radial unboundedness of V .
- By requirements 1 to 3, $x(t)$ goes to the target set C as $t \rightarrow \infty$ if $x(0)$ is sufficiently close to C .
- If C is bounded and requirement 4 holds then $x(t)$ goes to the target set C for any $x(0)$.

Extending the Lyapunov function

- Requirement 3 may be difficult to achieve.
- A Lyapunov function V only satisfying $\dot{V}(x) \leq 0$ for all $x \notin C$ may not guarantee that $x(t)$ tends to the set C as $t \rightarrow \infty$.
- However, further analysis can in certain circumstances reveal that $x(t)$ does go to C .
- If $\dot{V}(x) \leq 0$ for all $x \notin C$ but requirements 1 and 2, or 1, 2 and 4, are satisfied, it may still be possible to establish that $x(t)$ converges to C as $t \rightarrow \infty$ by the Krasovskii-LaSalle invariance principle.

Generalised Krasovskii-LaSalle invariance principle

- Let $V(x)$ satisfy requirements 1 and 2 for an extended Lyapunov function but only satisfies $\dot{V}(x) \leq 0$ for $x \notin C$. If C is bounded, let $\Omega_{r,c} = \{x \in \mathbb{R}^n \mid \|x - z\| \leq r \text{ for some } z \in C, V(x) \leq c\}$ ($r, c > 0$) be a set containing the target set, with r, c such that $\Omega_{r,c}$ is positively invariant. Otherwise, if C is unbounded, let $\Omega_{r,c} = \{x \in \mathbb{R}^n \mid \|x\| \leq r, V(x) \leq c\}$ be such that $\Omega_{r,c} \cap C \neq \emptyset$ and is positively invariant.

Then trajectories starting in $\Omega_{r,c}$ converge to the largest positively invariant set M in the set $I = \{x \in \Omega_{r,c} \mid \dot{V}(x) = 0\}$.

- If $M = \Omega_{r,c} \cap C$ then trajectories starting in $\Omega_{r,c}$ converge to M . If in addition C is bounded and the extended Lyapunov function meets requirement 4 (radial unboundedness) then all trajectories converge to C .

Example of convergence of trajectories to a set

- We begin by illustrating the application of extended Lyapunov functions to analyse the convergence of the trajectories of a system to a set (rather than a point)
- To do this we will need to invoke the generalised K-L invariance principle
- Consider a system described by the ODE

$$\begin{aligned}\dot{x}_1 &= x_2 - x_1^7(x_1^4 + 2x_2^2 - 10) \\ \dot{x}_2 &= -x_1^3 - 3x_2^5(x_1^4 + 2x_2^2 - 10)\end{aligned}\tag{1}$$

We want to analyse if there are trajectories of the system that converge to the set

$$C = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^4 + 2x_2^2 - 10 = 0\}$$

Example of convergence of trajectories to a set

- We consider $\Omega_{r,c} = \{x \in \mathbb{R}^2 \mid \|x - z\| \leq r, \text{ for some } z \in C, V(x) \leq c\}$ for some $r, c, > 0$ such that $\Omega_{r,c}$ is positively invariant, $C \subset \Omega_{r,c}$ and the origin is not contained in $\Omega_{r,c}$ (we will see shortly why), and the candidate Lyapunov function

$$V(x) = (x_1^4 + 2x_2^2 - 10)^2.$$

- We want to show that the largest positively invariant set M in $I = \{x \in \Omega_{r,c} \mid \dot{V}(x) = 0\}$ coincides with C .
- $\dot{V}(x) = -4(x_1^{10} + 3x_2^6)(x_1^4 + 2x_2^2 - 10) \leq 0$ and $\dot{V}(x) = 0 \Rightarrow x = (0, 0)$ or $x_1^4 + 2x_2^2 - 10 = 0$.
- Since the origin is not in $\Omega_{r,c}$ it is also not in I . So, we only need to look at the latter set of points that are all in C and $I = \{x \in \Omega_{r,c} \mid x \in C\}$.

Example of convergence of trajectories to a set

- Consider a trajectory starting in I at $t = 0$ with $x_1(0)^4 + 2x_2(0)^2 - 10 = 0$. To remain in I we must have

$$x_1(t)^4 + 2x_2(t)^2 - 10 = 0$$

$$\forall t > 0.$$

- By differentiating both sides of above with respect to t , the solution of the ODE satisfies

$$x_1(t)^3 \dot{x}_1(t) + x_2(t) \dot{x}_2(t) = 0, \quad \forall t \geq 0. \quad (2)$$

In a solution, the velocity $\dot{x}(t) = (x_1(t), x_2(t))^T$ is orthogonal to the vector $(x_1(t)^3, x_2(t))^T$ for all $t \geq 0$.

- Substituting $x_1(t)^4 + 2x_2(t)^2 - 10 = 0$ into the ODE gives,

$$\begin{aligned}\dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -x_1(t)^3.\end{aligned}\tag{3}$$

It is easy to check that the ODE (3) also satisfies (2). Thus solutions of (3) will trace out a curve satisfying $x_1(t)^4 + 2x_2(t)^2 - c = 0$ for some constant c , where $c = x_1(0)^4 + 2x_2(0)^2$.

- Indeed, integrating both sides of (2) with respect to time gives,

$$\begin{aligned}& \int_0^t (x_1(\tau)^3 \dot{x}_1(\tau) + x_2(\tau) \dot{x}_2(\tau)) d\tau = 0, \\ \Leftrightarrow & \int_0^t (x_1(\tau)^3 dx_1(\tau) + x_2(\tau) dx_2(\tau)) = 0 \\ \Leftrightarrow & \frac{1}{4}x_1(t)^4 + \frac{1}{2}x_2(t)^2 \\ & - \frac{1}{4}x_1(0)^4 - \frac{1}{2}x_2(0)^2 = 0, \quad \forall t \geq 0.\end{aligned}$$

Example of convergence of trajectories to a set

- Therefore, trajectories that remain in I when they start in I are those that start in C , i.e., starting at initial conditions satisfying $x_1(0)^4 + 2x_2(0)^2 = 10$.
- Since I does not contain the origin (again, this is why it was required that $\Omega_{r,c}$ does not contain the origin!), it follows that

$$M = \{x \in \Omega_{r,c} \mid x_1^4 + 2x_2^2 - 10 = 0\}.$$

- By the generalised K-L invariance principle all trajectories that start close to C (i.e., in the set $\Omega_{r,c}$) will converge to the set M .

- In fact, by replacing $\Omega_{r,c}$ with a different and larger positively invariant set and using the fact that the V is radially unbounded, it can be shown that all points not starting at the origin will converge to C (beyond what is covered in this course).

Design examples

- The Lyapunov based control system design method will be illustrated by its application in acceleration control of a vehicle, and control of mechanical systems.
- We will look at acceleration control of a vehicle to a single point and to a target set.

Acceleration control to a single point

We now move to consider a single vehicle motion control problem in $\mathbb{R}^3$. Each vector

$$x := (x_1, x_2, x_3)^T \in \mathbb{R}^3$$

generates the squared norm

$$\|x\|^2 = x_1^2 + x_2^2 + x_3^2,$$

whereas the scalar product $\langle x, y \rangle$ of two vectors $x = (x_1, x_2, x_3)^T$ and $y = (y_1, y_2, y_3)^T$ is defined by

$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

One can easily show that

$$\frac{d\|x(t)\|^2}{dt} = 2\langle x(t), \dot{x}(t) \rangle$$

and

$$\frac{d\|\dot{x}(t)\|^2}{dt} = 2\langle \dot{x}(t), \ddot{x}(t) \rangle$$

We denote by $x(t) = (x_1(t), x_2(t), x_3(t))^T$ the position of a vehicle in $\mathbb{R}^3$, i.e., $x_i(t), i = 1, 2, 3$ are the

coordinates of the position vector $x(t)$. Thus the velocity vector of the vehicle is

$$\dot{x}(t) = (\dot{x}_1(t), \dot{x}_2(t), \dot{x}_3(t))^T$$

and the acceleration vector of the vehicle is

$$\ddot{x}(t) = (\ddot{x}_1(t), \ddot{x}_2(t), \ddot{x}_3(t))^T.$$

Now, we will steer the vehicle by controlling acceleration, so the corresponding control system is

$$\ddot{x}(t) = u(t)$$

where $u(t) = (u_1(t), u_2(t), u_3(t))^T$ is the three dimensional (three degree of freedom) control input. The overall state of the system is

$$\xi(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \end{bmatrix},$$

and the overall controlled ODE is:

$$\dot{\xi}(t) = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} = \begin{bmatrix} \dot{x}(t) \\ u(t) \end{bmatrix},$$

with initial condition $\xi(0) = (x(0)^T, \dot{x}(0)^T)^T$.

The aim is to design a state-feedback control signal $u(x(t), \dot{x}(t))$ (depending on the position vector $x(t)$ and velocity vector $\dot{x}(t)$) that steers the vehicle to a target position $x_T \in \mathbb{R}^3$ (T denotes target). So here $C = \{(x, \dot{x}) \mid x = x_T, \dot{x} = 0\}$.

We use the candidate Lyapunov function

$$V(x, \dot{x}) = \frac{1}{2}(\|\dot{x}\|^2 + \|x - x_T\|^2)$$

then

$$\begin{aligned} \dot{V}(x(t), \dot{x}(t)) &:= \frac{dV(x(t), \dot{x}(t))}{dt} \\ &= \langle \dot{x}(t), \ddot{x}(t) \rangle + \langle x(t) - x_T, \dot{x}(t) \rangle \\ &= \langle \dot{x}(t), \ddot{x}(t) + x(t) - x_T \rangle \\ &= \langle \dot{x}(t), u(t) + x(t) - x_T \rangle \end{aligned}$$

Thus, in order to make $\dot{V}(x(t), \dot{x}(t)) \leq 0$, a natural choice of feedback control is

$$u(t) = -\kappa \dot{x}(t) - (x(t) - x_T), \quad \kappa > 0$$

Indeed, with such feedback control applied:

$$\dot{V}(x(t), \dot{x}(t)) = -\kappa \langle \dot{x}(t), \dot{x}(t) \rangle = -\kappa \|\dot{x}(t)\|^2 \leq 0.$$

Note that we only have $\dot{V}(x, \dot{x}) \leq 0$ outside of C rather than $\dot{V}(x, \dot{x}) < 0$ because $\dot{V}(x, \dot{x}) = 0$ at any point $x \neq x_T$ whenever $\dot{x} = 0$. Hence, we cannot conclude that $V(\xi(t)) \rightarrow 0$ and $\xi(t) \rightarrow (x_T, 0)$ as $t \rightarrow \infty$. However, we can check whether the Krasovskii-LaSalle invariance principle is applicable. Consider the set $\Omega_{r,c} = \{\xi \in \mathbb{R}^6 \mid \|\xi - (x_T, 0)\| \leq r, V(\xi) \leq c\}$, with $r, c > 0$ such $\Omega_{r,c}$ is positively invariant, and note that $\Omega_{r,c}$ contains the target point $(x_T, 0)$. Also, let $I = \{\xi \in \Omega_{r,c} \mid \dot{x} = 0\}$. We seek to determine all trajectories of the controlled system that remain in I when they start in I . We must have that if $\dot{x}(0) = 0$ then $\ddot{x}(t) = 0$ and $\dot{x}(t) = 0$ for all $t \geq 0$. But $\ddot{x}(t) = u(t) = -\kappa\dot{x}(t) - (x(t) - x_T)$, so we have $x(t) = x_T$ for all $t \geq 0$. Therefore, the largest invariant set M in I is just $M = \{(x_T, 0)\}$. Hence, the target $(x_T, 0)$ is asymptotically stable by the Krasovskii-LaSalle invariance principle. Since $V(\xi) > 0$ for all $\xi \neq (x_T, 0)$ and $V(\xi)$ is radially unbounded, $(x_T, 0)$ is globally asymptotically stable and $\xi(t)$ must converge to it as $t \rightarrow \infty$.

Acceleration control to a target set

Consider again a single vehicle with acceleration control as in the first design example but this time for motion on a plane rather than in 3D space. We now aim to do something a bit different. We would like to design a controller so that the vehicle comes to rest somewhere on a target set rather than a single point. Let $x = (x_1, x_2)^T$, $\dot{x} = (\dot{x}_1, \dot{x}_2)^T$, $u = (u_1, u_2)^T$ and the overall state be $\xi = (x_1, x_2, \dot{x}_1, \dot{x}_2)^T$. As before, we have acceleration control: $\ddot{x}_1 = u_1$ and $\ddot{x}_2 = u_2$.

The target set C_x for the vehicle position will be a circle with radius $R > 0$: $C_x = \{(x_1, x_2)^T \mid x_1^2 + x_2^2 = R^2\}$, while the actual target set C for the state $\xi \in \mathbb{R}^4$ is $C = \{(x_1, x_2, \dot{x}_1, \dot{x}_2)^T \mid x_1^2 + x_2^2 = R^2, \dot{x}_1 = \dot{x}_2 = 0\}$.

We then define the extended Lyapunov function

$$V(\xi) = \frac{1}{2}((\sqrt{x_1^2 + x_2^2} - R)^2 + \|\dot{x}\|^2).$$

Notice that $V(\xi) = 0$ if and only if $x \in C_x$ and $\dot{x} = 0$.

Control problem: Find a control signal $u(t) = \eta(x(t))$ such that $x(t) \rightarrow C_x$ and $\dot{x}(t) \rightarrow 0$ as $t \rightarrow \infty$.

We can compute $\dot{V}(\xi)$:

$$\begin{aligned}
\dot{V}(\xi) &= (\sqrt{x_1^2 + x_2^2} - R) \frac{x_1 \dot{x}_1 + x_2 \dot{x}_2}{\sqrt{x_1^2 + x_2^2}} + \langle \dot{x}, \ddot{x} \rangle, \\
&= (\sqrt{x_1^2 + x_2^2} - R) \frac{x_1 \dot{x}_1 + x_2 \dot{x}_2}{\sqrt{x_1^2 + x_2^2}} + \langle \dot{x}, u \rangle, \\
&= (\sqrt{x_1^2 + x_2^2} - R) \frac{x_1 \dot{x}_1 + x_2 \dot{x}_2}{\sqrt{x_1^2 + x_2^2}} + \dot{x}_1 u_1 \\
&\quad + \dot{x}_2 u_2 \\
&= \left(\frac{\sqrt{x_1^2 + x_2^2} - R}{\sqrt{x_1^2 + x_2^2}} x_1 + u_1 \right) \dot{x}_1 \\
&\quad + \left(\frac{\sqrt{x_1^2 + x_2^2} - R}{\sqrt{x_1^2 + x_2^2}} x_2 + u_2 \right) \dot{x}_2.
\end{aligned}$$

Now, when (x_1, x_2) is not at the origin, choose $u_j = -\kappa \dot{x}_j - \frac{\sqrt{x_1^2 + x_2^2} - R}{\sqrt{x_1^2 + x_2^2}} x_j$, with κ a positive constant, for $j = 1, 2$. We would then get that $\dot{V}(\xi)$ becomes

$$\dot{V}(\xi) = -\kappa(\dot{x}_1^2 + \dot{x}_2^2) = -\kappa \|\dot{x}\|^2.$$

However, we only have $\dot{V}(\xi) \leq 0$ outside of $C \cup \{(0, 0)\}$ rather than $\dot{V}(\xi) < 0$. This is because $\dot{V}(\xi) = 0$ for any ξ such that $\dot{x} = 0$, for any x (not necessarily $x \in C_x$). We again look to the Krasovskii-LaSalle's invariance principle.

Let $\Omega_{r,c} = \{\xi \in \mathbb{R}^4 \mid \|\xi - z\|^2 \leq r \text{ for some } z \in C, V(\xi) \leq c\}$ contain the target set C and is positively invariant with some $r < R$. Then $\Omega_{r,c}$ does not include the origin. Let $I = \{\xi \in \Omega_{r,c} \mid \dot{x} = 0\}$, we examine which trajectories under the control law chosen above remains in I when they start in I .

If $\dot{x}(0) = 0$ then we must have that $\ddot{x}(t) = 0$ and $\dot{x}(t) = 0$ for $t \geq 0$. But $\ddot{x}(t) = u(t)$, so we must have $u(t) = 0$ for all $t \geq 0$. Setting $u_1(t) = 0$ and $u_2(t) =$

0 for all $t \geq 0$ gives us either $\sqrt{x_1(t)^2 + x_2(t)^2} = R$ or $\frac{x_j(t)}{\sqrt{x_1(t)^2 + x_2(t)^2}} = 0$ for $j = 1, 2$. Since the origin is not contained in I , the second possibility is not feasible. For the first possibility, since we already have $\dot{x}(t) = 0$ for $t \geq 0$ for all trajectories that remain in I , the largest positively invariant set M in I is $M = C$. Thus, points starting in $\Omega_{r,c}$ will asymptotically converge to the set C under the chosen control law.

We could go further (beyond the purpose of this course). By replacing $\Omega_{r,c}$ with a different and larger positively invariant set and using the fact that the V is radially unbounded, it can be shown that all points not starting at the origin will converge to C .

Control of mechanical systems

There is a class of mechanical systems, called fully actuated mechanical systems, whose equation of motion can be expressed in the form

$$M(q)\ddot{q} + B(q, \dot{q})\dot{q} + G(q) = F$$

where $q \in \mathbb{R}^n$ is a vector of generalised coordinates for the system, and $F \in \mathbb{R}^n$ is the vector of input generalised forces. $M(q)$ is a symmetric positive definite matrix for each q . Also, typically $B(q, \dot{q}) + B(q, \dot{q})^T$ is a symmetric positive semidefinite matrix for each q and $\dot{q}$.

Suppose that we want to design the control $u = F$ to force $q(t) \rightarrow r$ as $t \rightarrow \infty$. Let us introduce the state

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

with $x_1 = q$ and $x_2 = \dot{q}$. Then we have

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -M(x_1)^{-1}(B(x_1, x_2)x_2 + G(x_1) - u) \end{aligned}$$

Define the Lyapunov function

$$V(x) = \frac{1}{2}x_2^T M(x_1)x_2 + \frac{\gamma_1}{2}(x_1 - r)^T(x_1 - r),$$

for a constant $\gamma_1 > 0$. Then $V(x) = 0$ only if $x_1 = r$ and $x_2 = 0$. We can straightforwardly compute $\dot{V}(x)$:

$$\begin{aligned}\dot{V}(x) &= \frac{1}{2}\dot{x}_2^T M(x_1)x_2 + \frac{1}{2}x_2^T M(x_1)\dot{x}_2 \\ &\quad + \frac{1}{2}x_2^T \frac{dM(x_1)}{dt}x_2 + \gamma_1 \langle \dot{x}_1, x_1 - r \rangle, \\ &= -x_2^T (B(x_1, x_2)x_2 - \frac{1}{2} \frac{dM(x_1)}{dt}x_2 + G(x_1) \\ &\quad - \gamma_1(x_1 - r) - u).\end{aligned}$$

If we choose u such that

$$-\frac{1}{2} \frac{dM(x_1)}{dt}x_2 + G(x_1) - \gamma_1(x_1 - r) - u = \gamma_2 x_2,$$

for some $\gamma_2 > 0$ then we would have

$$\dot{V}(x) = -\langle x_2, B(x_1, x_2)x_2 \rangle - \gamma_2 \|x_2\|^2.$$

If $B(x_1, x_2)$ is not symmetric, note that

$$x_2^T B(x_1, x_2)x_2 = x_2^T \tilde{B}(x_1, x_2)x_2, \text{ where } \tilde{B}(x_1, x_2)$$

is the symmetric matrix

$$\tilde{B}(x_1, x_2) = \frac{1}{2}(B(x_1, x_2) + B(x_1, x_2)^T).$$

This means that with

$$u = -\frac{1}{2} \frac{dM(x_1)}{dt} x_2 + G(x_1) - \gamma_1(x_1 - r) - \gamma_2 x_2$$

we have that $\dot{V}(x) \leq 0$. This is not enough to guarantee that $x_1(t) \rightarrow r$ since $\dot{V}(x)$ can vanish, even if $x_1 = q \neq r$, whenever $x_2 = \dot{q} = 0$. The system has to be examined further using the Krasovskii-LaSalle invariance principle to see if indeed $q(t) \rightarrow r$ as $t \rightarrow \infty$. In a tutorial exercise you will work out an example problem of this kind.